CSE 110: Computer Programming Sessional

Assignment: Grade Calculation Using Conditional Statements

Problem Statement

Write a C program that calculates the final mark, grade, and CGPA of a student for a course. The program will take attendance, class test marks, and term final marks as input.

The final mark will be calculated out of 100 using the following weight distribution:

Component	**Weight**
Attendance	10%
Class Test	20%
Term Final	70%

Mark Calculation Rules

Attendance: Take the total number of classes and the number of attended classes. First calculate the attendance percentage.

$$\text{Attendance Percentage} = \frac{\text{Attended Classes}}{\text{Total Classes}} \times 100$$

Then assign the attendance mark according to the following table:

Attendance Percentage	**Attendance Mark**
90% and above	10
85% to less than 90%	9
80% to less than 85%	8
75% to less than 80%	7
70% to less than 75%	6
65% to less than 70%	5
60% to less than 65%	4
Less than 60%	0

If the student gets 0 in attendance, the student is not eligible to participate in the final exam. In that case, the program must not take class test or term final marks as input. It should immediately print:

```
Not eligible for final exam.
```

Class Test: Take marks of 4 class tests. Each class test is out of 20. The best 3 marks will be counted. We will use `-1` as input to indicate that the student was absent in a class test.

$$\text{CT Mark} = \frac{\text{sum of best 3 CT marks}}{3}$$

Term Final: The term final has two sections: Section A and Section B. Each section has 4 questions, and each question is out of 10.

In each section, Question 1 is mandatory. From Question 2, Question 3, and Question 4, the student should answer only 2 questions. We will use `-1` as input to indicate that a question was not answered.

If a student answers more than 2 questions among Question 2, Question 3, and Question 4 in any section, the program should print a warning message for that section. However, for calculation, the program should still consider Question 1 and the best 2 marks among Question 2, Question 3, and Question 4.

For Section A, the warning message should be:

```
Warning: More than 2 optional questions answered in Section A.
```

For Section B, the warning message should be:

```
Warning: More than 2 optional questions answered in Section B.
```

For each section:

$$\text{Section Raw Mark} = Q1 + \text{best two marks among } Q2, Q3, Q4$$

Each section raw mark is out of 105 (Each question carrying 35 marks). It should be converted to 35 using:

$$\text{Section Final Mark} = \frac{\text{Section Raw Mark}}{105} \times 35$$

Therefore,

$$\text{Term Final Mark} = \text{Section A Final Mark} + \text{Section B Final Mark}$$

Finally,

$$\text{Final Mark} = \text{Attendance Mark} + \text{CT Mark} + \text{Term Final Mark}$$

Grading System

Use the following grading system:

Final Mark	**Grade**	**CGPA**
80 or above	A+	4.00
75 to less than 80	A	3.75
70 to less than 75	A-	3.50
65 to less than 70	B+	3.25
60 to less than 65	B	3.00
55 to less than 60	B-	2.75
50 to less than 55	C+	2.50
45 to less than 50	C	2.25
40 to less than 45	D	2.00
Less than 40	F	0.00

Input Instructions

The program should show appropriate messages while taking input. The program should first take attendance information. If the student is not eligible for the final exam, the program should terminate immediately.

For an eligible student, the input should be taken in the following order:

```
Enter total number of classes:
Enter number of classes attended:
Enter marks of 4 class tests:
Enter marks of 4 questions in Section A:
Enter marks of 4 questions in Section B:
```

Output Instructions

For an eligible student, the program should print the final mark, grade, and CGPA. The final mark should be printed up to 2 decimal places.

```
Final Mark: <final_mark>
Grade: <grade>
CGPA: <cgpa>
```

For a student who is not eligible for the final exam, the program should print:

```
Not eligible for final exam.
```

Sample Input and Output

Sample 1: Eligible Student

```
Enter total number of classes: 30
Enter number of classes attended: 27
Enter marks of 4 class tests: 15 18 -1 12
Enter marks of 4 questions in Section A: 28 25 -1 30
Enter marks of 4 questions in Section B: 26 29 31 -1

Final Mark: 81.33
Grade: A+
CGPA: 4.00
```

Sample 2: Warning for Extra Optional Question

```
Enter total number of classes: 30
Enter number of classes attended: 27
Enter marks of 4 class tests: 15 18 -1 12
Enter marks of 4 questions in Section A: 28 25 20 30
Enter marks of 4 questions in Section B: 26 29 31 -1

Warning: More than 2 optional questions answered in Section A.
Final Mark: 81.33
Grade: A+
CGPA: 4.00
```

Sample 3: Not Eligible Student

```
Enter total number of classes: 30
Enter number of classes attended: 17

Not eligible for final exam.
```

Important Condition

This assignment is designed to practice conditional statements. Therefore, students should mainly use conditional statements to solve the problem and avoid loops, arrays, or other topics that have not been covered yet.

Academic Integrity

Students must write the program by themselves. Copying code from classmates, seniors, online sources, or any other external source is strictly prohibited. Submitting code directly generated by ChatGPT or any other LLM or AI tool is also strictly prohibited.

Students may discuss the problem idea with others, but the submitted code must be their own work. Any plagiarism or direct use of LLM-generated code will result in **-100% marks** for this assignment.

Deadline

The assignment must be submitted by:

11 July 2026, 11:59 PM

This is a **strict deadline**.

Submission Instruction

Submit a single C source file using the following name format:

`student_id.c`

For example:

`2505001.c`